

# Deep Learning in Medicine: Predicting Surgical Skill Scores from Time Series Data

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## Abstract

Accurate assessment of surgical skill is essential for improving training outcomes and ensuring patient safety. Traditional assessment methods, however, are often subjective and resource-intensive. This project explores the use of deep learning techniques to predict surgical skill scores from time series data collected during simulation-based training. Using the Central Venous Catheterization (CVC) dataset by McGraw et al., we pretrain and adapt the InceptionTime architecture to extract meaningful representations of temporal features from raw sensor data. These features are then fed into a Gated Recurrent Unit (GRU) network to produce scalar skill score predictions for individual sessions. Our approach combines convolutional and recurrent neural architectures to capture both local patterns and long-range dependencies within the data. The results demonstrate the feasibility of using machine learning to model surgical learning curves and lay the groundwork for developing adaptive, personalized training systems in surgical education.

## Introduction

Surgery is a high-stakes discipline in which a surgeon's technical skill directly influences patient safety and clinical outcomes. However, proficiency levels among graduating surgeons-in-training can vary significantly; Some trainees exceed expectations while others fall short. This variability highlights the critical need for objective, accurate methods of performance assessment throughout surgical training programs [1].

Machine learning offers promising tools for analyzing the complex, dynamic nature of surgical performance. Mascagni et al. (2022) emphasize the potential of computer vision to enhance clinical decision-making and support safer surgical practices [4]. Moreover, the ability to predict individual learning curves could enable customizable training programs that optimize feedback, allocate resources efficiently, and ensure every trainee reaches expert-level proficiency by the end of training.

The primary goal of this research was to predict skill scores for surgeons-in-training at various stages of a simulation-based training program. These predicted scores serve as an intermediate step toward modeling the entire learning curve, an essential component in developing adaptive, personalized training systems.

We used data from McGraw et al.'s simulation-based curriculum for ultrasound-guided central venous catheterization (CVC), to test our models for feature extraction and prediction of skill scores [3].

To extract meaningful representations from the raw time series data, we selected InceptionTime, a deep learning model originally designed for time series classification (TSC), due to its strong performance in benchmark studies [2]. Unlike standard CNNs such as AlexNet, which require two-dimensional inputs, InceptionTime is optimized for one-dimensional sequential data. However, since pretrained weights were not publicly available, we first pretrained the model on a dataset similar in structure to our CVC data.

Because time series data is inherently sequential, the modified InceptionTime model outputs a sequence of feature vectors over time. These were then passed into a separate prediction model designed to produce a single scalar output representing the predicted skill score for each session.

For this prediction stage, we adopted a many-to-one Recurrent Neural Network (RNN) architecture. While a standard RNN was initially considered, its vulnerability to vanishing gradients during backpropagation made it unsuitable for long sequences. Instead, we selected the Gated Recurrent Unit (GRU), a variant of the RNN that incorporates gating mechanisms to preserve memory over longer time steps and mitigate the vanishing gradient issue [5].

The following sections of this paper detail the implementation process for this project, the model architecture, the pretraining strategy, and the experimental results on the CVC simulation dataset.

## Prior Work

The interest in using time series data as a tool in surgical skill assessment has grown significantly with the development of advanced machine learning models. In particular, recurrent neural networks (RNNs) and convolutional neural networks (CNNs) have been used to assess technical skill from motion data and video sequences. A systematic review of machine learning approaches for technical skill assessment in surgery highlighted the increasing use of data-driven methods in evaluating surgical performance, particularly those relying on kinematic time series and sensor-based features [1].

Computer vision has also emerged as a promising area in surgical education, enabling automated assessment from video recordings and offering new opportunities for real-time feedback [4]. Architectures such as AlexNet demonstrated the potential of deep CNNs for image and video analysis, though these approaches require large amounts of labeled data and often lack temporal resolution [2]. Other approaches, such as InceptionTime by H.Fawaz et al., address time series data directly, taking one-dimensional tensor data as input rather than the two-dimensional inputs expected by AlexNet for image classification [2].

To address limitations in standard surgical-training-assessment methods, McGraw et al. developed and evaluated a simulation-based curriculum for ultrasound-guided central venous catheterization (CVC),

where skill was rated by experts using methods such as Global Rating Scale (GRS) across multiple training sessions [3].

Building upon these works, this project aims to combine convolutional and recurrent architectures to extract meaningful temporal features and predict skill scores for surgeons-in-training, contributing toward the long-term goal of modeling surgical learning curves.

## Methods

### 1. Dataset Introduction

To predict skill scores effectively, we utilized a dataset documenting multiple time points throughout the surgical training program. The dataset used in this study originates from the work by McGraw et al., described in the paper “*Development and Evaluation of a Simulation-Based Curriculum for Ultrasound-Guided Venous Catheterization*” [3]. This dataset focuses on examining the amount of study and practice required to achieve technical proficiency in ultrasound-guided central venous catheter (CVC) insertion, a critical skill in surgical practice [3].

The CVC time series data is collected from trainees at four separate training sessions: 2015 tracking data, Session 1 (2016), Session 2 (2016), and Session 3 (2016). For each participant, performance data was recorded both at the beginning and the end of each session, providing a total of eight time points per participant. The nature of this temporal resolution enables modeling of multiple skill scores per session, facilitating future work to predict entire learning curves.

Recorded data consists of multi-dimensional time series signals captured via motion sensors placed on the left and right hands of every participant during simulated CVC insertion procedures. The sensory data is represented as sixteen-element rotational matrices, with the first twelve elements corresponding to sensor-acquired values. The motion-sensor data captures the left and right hand movements of the trainees, correlating to their technique, precision, and efficiency. Additionally, as the data was collected as part of a structured simulation-based curriculum, it offers a controlled environment suitable for studying skill progression.

To benchmark trainee performance against expert-level standards, ground-truth skill scores generated by Associate Professor Matthew S. Holden were used. These scores were computed using Python scripts available in the *skill\_scores* file on our GitHub repository. The ground truth skill scores were passed as the target in the GRU model, allowing analysis of loss metrics (mean squared error, mean absolute error) and evaluation of the model’s ability to predict participants’ skill scores across the four training sessions.

### 2. Modifying InceptionTime for Feature Extraction

The original InceptionTime architecture, developed by H.Fawaz et al., is a deep neural network inspired by Inception modules from computer vision, specifically adapted for time series data [2]. Often described as the “AlexNet for time series classification”, it was an optimal choice for our feature extraction task. Our approach involved modifying InceptionTime to be a feature extractor, pretraining its weights on a

separate dataset, running this modified model on the CVC dataset to extract features, and finally feeding those extracted features into a Gated Recurrent Unit (GRU) for skill score prediction.

The original InceptionTime outputs class probabilities via a Global Average Pooling (GAP) layer followed by a fully connected softmax layer [2]. However, these outputs represent final class predictions rather than the rich intermediate feature representations required for predictions via GRU. To adapt InceptionTime as a feature extractor, we created a separate model that outputs the intermediate feature tensor immediately before the GAP layer. We replicated the Inception block stack but excluded the final GAP and dense layers. This modified model produces multi-channel feature sequences with preserved temporal structure that can be passed to a GRU model for prediction. The modification was implemented in the `build_feature_extractor()` function in the `Classifier_INCEPTION` class, which returns a model mapping inputs to the convolutional output just before the global pooling step.

### 3. Pretraining on MotionSenseHAR Data

Since pretrained InceptionTime weights are not publicly available, and training from scratch on the small CVC dataset may lead to overfitting, it was necessary to find a larger and related dataset for pretraining the modified InceptionTime feature extractor.

The CVC data is structured as sixteen-element rotational matrices for both left hand and right hand data, with the last four values always fixed at '0 0 0 1'. Thus, only the first twelve contain relevant data, resulting in a twelve-channel multivariate time series. To pretrain on a dataset with a similar structure, we prioritized finding a multivariate time series dataset with twelve channels. While an alternative approach could have involved pretraining on a univariate dataset and then applying the feature extractor separately to each CVC channel, this was far less efficient.

We therefore selected the MotionSenseHAR dataset from the UCR archive, which contains twelve channels, matching the dimensionality of the CVC data.. The similarity in both data type (motion sensor data) and channel dimensionality allows the model to learn and extract transferable, meaningful features for the GRU downstream task.

### 4. Feature Extraction on CVC data

Using the pretrained weights from the modified InceptionTime model (pretrained on MotionSenseHAR data), we developed a feature extractor tailored for the CVC dataset. Documentation for `extract_cvc_features.py` is available in the wiki in our GitHub repository.

The CVC feature extraction script uses manually specified paths for pretrained weights, datasets, and output folders. It resamples the raw sensor data to a fixed number of time steps (e.g., 250) to shorten sequence length for processing and then extracts the features from both left and right hand data by loading the pretrained InceptionTime model, and calling `build_feature_extract()` to construct a model that returns intermediate convolutional features before the final pooling and dense layers. Extracted feature representations are saved as binary `.npy` files and organized into subfolders corresponding to the four

CVC training sessions. The organization method was meant to keep track of all data and align it accordingly with the generated ground-truth skill scores when predicting using the GRU model.

## 5. Computing Predictions with a Gated Recurrent Unit (GRU)

The vanishing gradient problem motivated the choice of a GRU model over a standard RNN for prediction. Standard RNNs tend to suffer from exponentially diminishing gradients during training, which hampers their ability to retain long-term dependencies in sequential data. GRUs mitigate this issue by incorporating gating mechanisms that maintain a memory cell, allowing relevant information to be preserved across long sequences [5].

In our workflow, before feeding features into the GRU, we first load and concatenate the separately extracted left and right hand feature sets from their respective .npy files. The ground-truth skill scores, which serve as the prediction targets, are loaded from a .tsv file. These targets are split into four session-based subsets mirroring the organization of the CVC data. The `load_targets()` function facilitates loading the target values from the session-organized subfolders, which aligns with the feature data's folder structure. This organization enables a dataset split of approximately 60% training data (2015 data and 2016 Session 1), 20% validation data (2016 Session 2), and 20% testing data (2016 Session 3). This split strategy helps prevent data leakage by keeping validation and test sets separate, thus reserving them for unbiased evaluation and model progress assessment [5].

The GRU model is constructed using the `build_gru_model()` function, which accepts the input shape as a parameter. After extensive hyperparameter tuning, the optimal architecture consists of four stacked GRU layers, with the number of neurons starting at 100 and decreasing by 25% after each layer. To combat overfitting, a dropout layer with a fixed rate of 0.3 follows each GRU layer. Finally, a single dense layer outputs a scalar value, consistent with the many-to-one sequence prediction format required for this task.

For compilation, the model uses mean squared error (MSE) as the loss function and mean absolute error (MAE) as a secondary metric for performance evaluation and hyperparameter tuning. The Adam optimizer is employed with a learning rate of 0.001, both of which are standard choices in machine learning practice.

Model training is performed via `model.fit()` on the training dataset (2015 and 2016 Session 1), with matching target labels organized in corresponding subfolders (`targets_2015` and `targets_2016_Session1`). The training runs for 15 epochs with a batch size of 64. Validation data is set to the 2016 Session 2 subset and its corresponding targets.

After training, `model.predict()` is called on the training, validation, and test datasets to generate the final skill-score predictions.

## 6. Calculating and Interpreting Loss: Metrics and Zero-rule Regressor

To thoroughly evaluate model performance, we used two loss metrics alongside extensive validation methods. We implemented a zero-rule regressor as a simple baseline, which predicts the mean of the target values for all inputs. This baseline allows us to assess whether our GRU model performs better than simply guessing the average skill score.

The evaluation metrics used were Mean Squared Error (MSE) and Mean Absolute Error (MAE). These metrics were calculated for both the zero-rule regressor and the trained GRU model. By comparing these values for each metric, we could quantitatively demonstrate that the GRU model’s performance improved as hyperparameter tuning progressed, outperforming the baseline average prediction.

## Experiments

| Exp # | Batch | GRU Layers and Size             | Dropout | Epochs | Test MAE      | Test MSE      |
|-------|-------|---------------------------------|---------|--------|---------------|---------------|
| 1     | 32    | L1: 10                          | 0.3     | 10     | 0.9295        | 1.0798        |
| 2     | 32    | L1: 50                          | 0.3     | 20     | 0.7586        | 0.6836        |
| 3     | 64    | L1: 50, L2: 50                  | 0.3     | 15     | 0.5196        | 0.4166        |
| 4     | 64    | L1: 100, L2: 50, L3: 25         | 0.3     | 15     | 0.5096        | 0.3158        |
| 5     | 64    | L1: 100, L2: 75, L3: 50, L4: 25 | 0.4     | 15     | 0.6750        | 0.5662        |
| 6     | 64    | L1: 100, L2: 75, L3: 50, L4: 25 | 0.3     | 15     | <b>0.1447</b> | <b>0.0357</b> |

**Table 1.** GRU hyperparameter tuning experiments. Test MAE and Test MSE are reported for each experiment and compared against the baseline performance of a zero-rule regressor (ZRR), which achieved an MAE of **0.6716** and MSE of **0.5002**.

### 1. Analyzing Loss

To analyze and evaluate the model’s loss, we implemented a zero-rule regressor on the ground-truth skill scores. This baseline model always predicts a fixed value: the average skill score across the data. Since the zero-rule regressor represents a simple average prediction, our goal is for the GRU model’s loss to be significantly lower than that baseline. As the loss values for our model decrease, it indicates improved performance.

### 2. Tuning Hyperparameters

We conducted a series of experiments to optimize the GRU model’s architecture and training parameters, as summarized in Table 1. Initially, with a single GRU layer of 10 neurons, a batch size of 32, and 10 epochs (Experiment 1), the model underperformed relative to the zero-rule regressor baseline, indicating insufficient capacity for the task. Increasing the number of neurons in the single layer to 50 and extending

training to 20 epochs (Experiment 2) improved performance, reducing both MAE and MSE closer to, but still above, the baseline.

Adding a second GRU layer of equal size (50 neurons each) and adjusting the batch size to 64 while training for 15 epochs (Experiment 3) further enhanced results, with a noticeable decrease in loss metrics. Building on this, the introduction of a third layer with progressively decreasing neuron counts (100, 50, and 25 neurons respectively; Experiment 4), led to even better performance, confirming that deeper architectures with decreasing layer sizes capture hierarchical temporal features more effectively.

However, increasing dropout from 0.3 to 0.4 (Experiment 5) resulted in degraded performance, highlighting that excessive regularization can harm learning. Maintaining a dropout rate of 0.3 after each GRU layer (Experiment 6), combined with four layers of decreasing neurons (100, 75, 50, and 25), batch size 64, and 15 epochs, yielded the best results by a significant margin, achieving a Test MAE of approximately 0.145 and Test MSE of 0.036, outperforming the zero-rule regressor baseline significantly.

These findings highlight the importance of model depth, neuron distribution, dropout tuning, and epoch selection in improving predictive accuracy for skill score regression using GRU models.

## Results

The performance of the GRU model in predicting skill scores was evaluated using mean absolute error (MAE) and mean squared error (MSE) as metrics, comparing it against a zero-rule regressor baseline that predicts the average score for all samples. The model outputs a scalar skill score prediction for each data sample, and the reported loss metrics represent averages over the test set.

Initial experiments with a single GRU layer of 10 neurons (batch size 32, 10 epochs) resulted in worse performance than the baseline, indicating insufficient model capacity for this task. Increasing the number of neurons and adding layers progressively improved the results. Notably, a four-layer GRU model with neuron counts decreasing from 100 to 25 per layer, a batch size of 64, dropout of 0.3 after each layer, and 15 training epochs achieved the best performance, with test MAE of approximately 0.145 and test MSE around 0.036. This represented a substantial improvement over the zero-rule regressor baseline (MAE  $\approx$  0.672, MSE  $\approx$  0.5), demonstrating that the model successfully learned meaningful temporal representations from the feature data.

We also observed that increasing dropout beyond 0.3 negatively affected performance, and that selecting an intermediate number of epochs helped avoid both underfitting and overfitting. These results indicate that the tuned GRU architecture is effective for skill score prediction within this domain and dataset.

## Discussion

The main objective of this study was to develop a tool capable of predicting skill scores, with the aim of supporting downstream projects focused on modeling the full learning curves of surgeons-in-training and

enabling customizable training programs. The feature extractor and GRU predictor were designed and optimized to analyze and improve predictive performance.

As shown in Table 1, model performance improved steadily as hyperparameters were tuned. Given the twelve-week timeline of the internship during which this work was developed, we stopped tuning after Experiment 6 to retain a working, well-performing model within the project constraints.

The achieved test MAE of approximately 0.145 suggests the model predicts skill scores with reasonable accuracy, making it a promising tool for integration into adaptive training frameworks. However, the model's performance and generalizability may be limited by the relatively small size of the CVC dataset used for training and evaluation.

Further work is needed to assess the model's generalization capabilities across larger and more diverse datasets, as well as to explore the impact of other hyperparameters, such as the number of time steps considered during feature extraction. Future research might also investigate alternative recurrent architectures, such as LSTMs or transformer-based models, and perform more extensive hyperparameter tuning to further enhance prediction accuracy.

## 1. Limitations

Due to the limited duration of the internship, only six hyperparameter tuning experiments were conducted, which restricts the ability to fully explore the parameter space and understand the influence of individual hyperparameters on model performance. Additional tuning may reveal more nuanced effects on MAE and MSE metrics and identify more optimal model configurations.

The small dataset size also limits the generalizability of the findings. Applying the GRU model to larger datasets may require retraining or reconfiguring architectural elements such as the number and placement of dropout layers. Additionally, decisions regarding the number of time steps used in feature extraction were made based on prior trials with a different dataset (MotionSenseHAR from the UCR archive), as specific experiments on this aspect were limited for the current dataset.

## 2. Future Work

This task is part of a larger goal to predict the full learning curve of a surgeon-in-training using data collected from only a few initial training sessions. The outputs from the developed GRU model, which leverages a modified feature extractor based on H. Fawaz et al. 's InceptionTime architecture, can serve as inputs for downstream models that construct a continuous learning curve over multiple sessions rather than providing isolated skill score predictions for individual sessions.

Future work could explore integrating temporal progression models, such as sequence-to-sequence networks, to better capture the trajectory of skill acquisition across time. Additionally, expanding the

dataset to include a broader range of surgical tasks and trainee profiles would improve the robustness and generalizability of the learning curve predictions.

There is also potential for incorporating multimodal data sources, such as video, kinematic data, and physiological signals, to enrich the feature space and further enhance prediction accuracy. Exploring alternative recurrent architectures, attention mechanisms, or transformer-based models may yield improvements in capturing long-term dependencies and subtle changes in performance.

Finally, user studies involving surgical trainers and trainees could help validate the practical utility of the predictive models in real-world training scenarios, ultimately enabling personalized and adaptive surgical education programs.

## Conclusion

In this study, we developed and optimized a GRU-based model for predicting skill scores from surgical training data, demonstrating a substantial improvement over a zero-rule baseline. Our experiments underscore the critical role of hyperparameter tuning, model depth, and dropout regularization in achieving optimal performance.

While limitations related to dataset size and experimental scope remain, the model provides a foundation for future work in surgical skill assessment. Potential directions include expanding dataset diversity, exploring more advanced sequence modeling techniques, and integrating various data types to better capture the complex dynamics of surgical learning. Ultimately, these efforts aim to enable personalized training programs that adapt to individual learning trajectories and enhance outcomes in surgical education.

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Additionally, this work builds on the InceptionTime architecture by H. Fawaz et al., and we gratefully acknowledge their contribution to the time series deep learning community.

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